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Inventor(s)	Kim; Junho

Clothes treating apparatus

Abstract

A clothes treating apparatus including: a cabinet; a tub provided in the cabinet and configured to collect wash water; a circulation device configured to circulate the wash water contained in the tub and including a circulation pump, a first circulation path to guide the wash water from the tub to the circulation pump, and a second circulation path to guide the wash water from the circulation pump to the tub so that the wash water circulates by flowing from the tub to the circulation pump through the first circulation path and flowing back to the tub from the circulation pump through the second circulation path; and a detergent supply device configured to supply a detergent to the circulation device and including a detergent case, and a detergent pump connected to the detergent case and mounted on the first circulation path to supply the stored detergent through the first circulation path.

Inventors:	Kim; Junho (Suwon-si, KR)
Applicant:	SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)
Family ID:	86992441
Assignee:	SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)
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Primary Examiner: Perrin; Joseph L.

Attorney, Agent or Firm: STAAS & HALSEY LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation application, filed under 35 U.S.C. § 111(a), of International Application PCT/KR2022/014577 filed Sep. 28, 2022, and is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0000244, filed on Jan. 3, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(1) The present disclosure relates to a clothes treating apparatus, and more particularly, to a clothes treating apparatus including a detergent supply device.

2. Description of the Related Art

(2) In general, a clothes treating apparatus includes a washing machine configured to wash laundry loaded in a rotary tub by mutual friction by rotating the laundry and a detergent together using a driving force of a motor, a dryer configured to dry an object to be dried and contained in the rotary tub by rotating the rotary tub, and the like.

(3) A clothes treating apparatus may include a detergent supply device provided in a cabinet, and the detergent supply device is a device configured to supply a detergent into a tub. The detergent supply device may include a detergent case, and a detergent pump configured to supply the detergent contained in the detergent case into the tub in response to docking with the detergent case.

(4) In this case, the detergent discharged from the detergent pump should be mixed with wash

water and the detergent may remain in a flow path from the detergent pump to a space in which the detergent is to be mixed with wash water. In the case where the detergent residues are left, contamination and solidification of the detergent may occur.

SUMMARY

(5) In accordance with an aspect of the present disclosure, a clothes treating apparatus includes: a cabinet; a tub provided in the cabinet and configured to collect wash water; a circulation device configured to circulate the wash water contained in the tub, and the circulation device including a circulation pump, a first circulation path provided to guide wash water from the tub to the circulation pump, and a second circulation path provided to guide wash water from the circulation pump to the tub so that the wash water circulates by flowing from the tub to the circulation pump through the first circulation path and flowing back to the tub from the circulation pump through the second circulation path; and a detergent supply device configured to supply a detergent to the circulation device, and the detergent supply device including a detergent case, and a detergent pump connected to the detergent case to store the detergent and mounted on the first circulation path to supply the stored detergent through the first circulation path.

(6) The first circulation path may include: a drain path to guide the wash water discharged from the tub out of the tub; and a mixing path to guide the wash water to the circulation pump flowing from the drain path and mounted with the detergent pump.

(7) The drain path may include: a first drain path to guide the wash water from the tub out of the tub; and a second drain path connected to the first drain path and branching from the first drain path.

(8) The clothes treating may further include a drain pump configured to receive the wash water from the first drain path and mounted on the second drain path.

(9) The clothes treating apparatus may further include a water supply device configured to supply the wash water into the tub from an external water supply source, wherein the wash water flowing into the tub by the water supply device flows to the tub after passing through the first circulation path, the circulation pump and the second circulation path in response to the circulating of the wash water by the circulation pump.

(10) The clothes treating apparatus may further include a processor configured to selectively operate the circulation pump or the drain pump.

(11) The processor may be further configured to operate the circulation pump in response to determining that a water level in the tub reaches a predetermined water level.

(12) The processor may be further configured to operate the detergent pump in response to the operating of the circulation pump.

(13) The detergent case may include: a first detergent case; and a second detergent case separated from the first detergent case in a lateral direction, wherein the processor is further configured to operate the circulation pump based on initiation of a washing mode, the detergent pump to allow a detergent contained in the first detergent case to flow into the detergent pump based on the operation of the circulation pump, and the detergent pump to allow a detergent contained in the second detergent case to flow into the detergent pump based on initiation of a rinsing mode.

(14) The processor may be further configured to turn off the circulation pump and turn on the drain pump to drain the wash water contained in the tub.

(15) The detergent supply device and the circulation device may be disposed below the tub.

(16) The detergent supply device may be a first detergent supply device, and the clothes treating apparatus may further include a second detergent supply device disposed above the tub to supply the detergent into the tub.

(17) The clothes treating apparatus may further include: a fan above the tub to form a circulating air flow passing through the tub; and a heat exchanger above the tub to dehumidify and heat air discharged from the tub and supply the dehumidified and heated air to the tub.

(18) The detergent supply device may further include a drawer configured to accommodate the

detergent case and to be withdrawn from or inserted into the cabinet.

(19) The drawer may include: a first drawer configured to accommodate the detergent case and withdrawn from or inserted into the cabinet; and a second drawer coupled to a side wall of the cabinet, configured to accommodate the first drawer, and slidably coupled to the first drawer allowing the first drawer to slide.

(20) A clothes treating apparatus according to an embodiment of the present disclosure may include a cabinet, a tub in the cabinet, a water supply device configured to supply wash water into the tub from an external water source, and a detergent supply device configured to supply a detergent into the tub, disposed at a lower portion of the cabinet, the detergent supply device comprising a detergent case, a detergent pump including an inlet through which the detergent is introduced from the detergent case and an outlet through which the introduced detergent is to discharge the detergent, and a mixing space formed therein to receive wash water from the tub and directly connected to the outlet of the detergent pump to receive the discharged detergent from the detergent pump.

(21) The supplied wash water from the external water supply source may flow back to the tub after passing through the water supply device, the tub, and the mixing space.

(22) A clothes treating apparatus according to an embodiment of the present disclosure may include a cabinet, a tub in the cabinet and configured to collect wash water, a detergent supply device configured to supply a detergent into the tub, the detergent supply device including a detergent case to accommodate the detergent, a detergent pump including an inlet through which the detergent is introduced from the detergent case and an outlet through which the introduced detergent is discharged to discharge the detergent, and a mixing space formed therein to receive wash water from the tub and directly connected to the outlet of the detergent pump to receive the detergent from the detergent pump, a drain pump connected to the tub to drain the wash water contained in the tub out of the cabinet, a circulation pump connected to the tub to circulate the wash water, and a processor configured to selectively operate the drain pump or the circulation pump.

(23) The processor may be further configured to operate the detergent pump in response to the operating of the circulation pump.

(24) The processor may be further configured to turn off the circulation pump and turn on the drain pump to drain the wash water contained in the tub.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a clothes treating apparatus according to an embodiment of the present disclosure.

(2) FIG. 2 is a side cross-sectional view of the clothes treating apparatus illustrated in FIG. 1.

(3) FIG. 3 is a side cross-sectional view of the clothes treating apparatus illustrated in FIG. 1.

(4) FIG. 4 is a perspective view showing a state where a detergent supply device is withdrawn from a cabinet in the clothes treating apparatus illustrated in FIG. 1.

(5) FIG. 5 is a perspective view showing a state where the cabinet is omitted from the clothes treating apparatus illustrated in FIG. 1.

(6) FIG. 6 is a perspective view of the clothes treating apparatus illustrated in FIG. 5.

(7) FIG. 7 is a block diagram schematically illustrating the clothes treating apparatus illustrated in FIG. 1.

(8) FIG. 8 is a perspective view of the detergent supply device and the circulation device of the clothes treating apparatus illustrated in FIG. 1.

(9) FIG. 9 is an exploded perspective view of the detergent supply device and the circulation device illustrated in FIG. 8.

- (10) FIG. 10 is an exploded perspective view of the detergent pump illustrated in FIG. 9.
- (11) FIG. 11 is a cross-sectional view of the detergent supply device illustrated in FIG. 8.
- (12) FIG. 12 is a cross-sectional view of the detergent supply device illustrated in FIG. 8.
- (13) FIG. 13 is a control block diagram of the clothes treating apparatus illustrated in FIG. 1.
- (14) FIG. 14 is a control flow chart of the clothes treating apparatus illustrated in FIG. 1.

DETAILED DESCRIPTION

(15) Reference will now be made in detail to the embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. The embodiments described in the specification and shown in the drawings are only illustrative and are not intended to represent all aspects of the invention, such that various modifications may be made without departing from the spirit of the invention.

(16) In the drawings, like reference numerals denote like elements or components having substantially same functions.

(17) Throughout the specification, the terms used are merely used to describe particular embodiments, and are not intended to limit the present disclosure. As used herein, the singular forms are intended to include the plural forms as well, unless otherwise stated. Also, it is to be understood that the terms such as “include”, “have”, or the like, are intended to indicate the existence of the features, numbers, operations, components, parts, or combinations thereof disclosed in the specification, and are not intended to preclude the possibility that one or more other features, numbers, operations, components, parts, or combinations thereof may exist or may be added.

(18) It will be understood that, although the terms “first”, “second”, etc., may be used herein to describe various elements, these elements should not be limited by these terms. The above terms are used only to distinguish one component from another. For example, a first component discussed below could be termed a second component, and similarly, the second component may be termed the first component without departing from the teachings of this disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(19) The terms used throughout the specification “front”, “rear”, “left”, and “right”, and the like are defined based on the drawings and the shape and position of each element are not limited by these terms. A clothes treating apparatus 1 according to various embodiments of the present disclosure may include washing machines, clothes dryers, clothes care apparatuses, and the like. However, for descriptive convenience, the embodiments of the present disclosure will be described below using a front-loading washing machine 1 as an example.

(20) In addition, a detergent may include not only laundry detergents but also rinses, deodorizers, disinfectants, or fragrances for dryers.

(21) A clothes treating apparatus may be capable of preventing detergent residues in a detergent supply device.

(22) A clothes treating apparatus may be capable of reducing contamination and solidification of a detergent by preventing detergent residues.

(23) A clothes treating apparatus may have improved user convenience.

(24) According to an embodiment of the present disclosure, a clothes treating apparatus capable of preventing detergent residues by mounting a detergent pump on a circulation path may be provided.

(25) According to an embodiment of the present disclosure, a clothes treating apparatus in which a detergent pump is directly connected to a circulation path may be provided.

(26) According to an embodiment of the present disclosure, a clothes treating apparatus having improved user convenience by preventing malfunctioning caused by detergent residues in advance may be provided.

(27) Hereinafter, the embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

(28) FIG. 1 is a perspective view of a clothes treating apparatus according to an embodiment of the

present disclosure. FIG. 2 is a side cross-sectional view of the clothes treating apparatus illustrated in FIG. 1. FIG. 3 is a side cross-sectional view of the clothes treating apparatus illustrated in FIG. 1. FIG. 4 is a perspective view showing a state where a detergent supply device is withdrawn from a cabinet in the clothes treating apparatus illustrated in FIG. 1.

(29) Referring to FIGS. 1 to 4, a washing machine **1** includes a cabinet **10** defining an exterior appearance of the washing machine **1** and accommodating various components, a tub **20** provided in the cabinet **10** to contain mixed water in which wash water and a detergent are mixed, a drum **30** provided in the tub **20** to contain laundry and configured to rotate, and a drive device **60** configured to rotate the drum **30**. The drive device **60** is located behind the tub **20** and configured to rotate the drum **30** in the forward-reverse directions to perform washing, rinsing, spin-drying, and drying operations.

(30) The cabinet **10** may be formed in a substantially hexahedral shape. The cabinet **10** may be formed in a box shape with an open front. The cabinet **10** may include a rear plate **12d** and both side plates **12b** and **12c** located at both sides of the rear plate **12d** in the forward direction. Although the cabinet **10** including the rear plate **12d** and the side plates **12b** and **12c** which are integrated with each other is shown herein by way of example, the present disclosure is not limited thereto. The cabinet **10** may include a front panel **11** provided in the open front, a base **19** provided at the bottom, and a top cover **12e** provided at the upper surface.

(31) The front panel **11** may include a first front panel **11a** and a second front panel **11b** located at an upper side of the first front panel **11a**.

(32) The first front panel **11a** may be provided in front of the drum **30**. The first front panel **11a** may be located at a lower side of the second front panel **11b**. The first front panel **11a** may have an opening **13** through which laundry is loaded into the drum **30**. The opening **13** of the first front panel **11a** is formed to allow the laundry to be loaded or unloaded in the forward direction of the cabinet **10**. The opening **13** of the first front panel **11a** may be open or closed by a door **40**. The door **40** may include a door frame **41** and a door glass **42**. The door glass **42** may be formed of a transparent tempered glass material such that the interior of the cabinet **10** is visible therethrough.

(33) The second front panel **11b** may be provided at an upper-front side of the cabinet **10**. The second front panel **11b** may be disposed on the upper side of the first front panel **11a**. The upper end of the second front panel **11b** is provided to be coupled to the top cover **12e**.

(34) The second front panel **11b** may include a control panel (not shown) including an inputter configured to receive an operation command from a user and a display (not shown) configured to display operation information of the washing machine **1**.

(35) The base **19** may be configured to cover the bottom of the cabinet **10**.

(36) A water supply device **14** may be provided above the tub **20**. The water supply device **14** may include a water supply valve **14b** configured to control water supply and water supply pipes **14a**. The water supply device **14** may supply water into the tub **20**. For example, water flowing into the cabinet **10** through the water supply device **14** may be collected in the cabinet **10**.

(37) The clothes treating apparatus according to an embodiment of the present disclosure may include detergent supply devices **90** and **100**. The detergent supply devices **90** and **100** may supply detergents into the tub **20**. The detergent supply devices **90** and **100** may be provided in plural. The plurality of detergent supply devices **90** and **100** may include a first detergent supply device **100** and a second detergent supply device **90**.

(38) The first detergent supply device **100** may be disposed below the tub **20**. For example, the first detergent supply device **100** may be disposed at a lower portion in the cabinet **10**. The first detergent supply device **100** may be disposed at the bottom in the cabinet **10**. The first detergent supply device **100** may be withdrawn from the cabinet **10** or inserted into the cabinet **10** at a lower portion of the front panel. The first detergent supply device **100** may be slidably moved (See FIG. 4).

(39) The first detergent supply device **100** may allow the detergent to flow into the tub **20**. For

example, the first detergent supply device **100** may allow a liquid laundry detergent and/or a fabric softener to flow into the tub **20**. However, types of the detergent are not limited thereto.

(40) The second detergent supply device **90** may be disposed above the tub **20**. For example, the second detergent supply device **90** may be disposed at an upper portion in the cabinet **10**. The second detergent supply device **90** may be disposed at the top in the cabinet **10**. The second detergent supply device **90** may be installed in the second front panel **11b**. The second detergent supply device **90** may be withdrawn from the cabinet **10** or inserted into the cabinet **10** at an upper portion of the front panel **11**. The second detergent supply device **90** may be slidably moved.

(41) The second detergent supply device **90** may allow the detergent to flow into the tub **20**. For example, the second detergent supply device **90** may allow a solid laundry detergent and/or fabric softener to flow into the tub **20**. However, types of the detergent are not limited thereto.

(42) The first detergent supply device **100** may be disposed below the tub **20** and supply the detergent into the tub **20** during a water supplying process, and the second detergent supply device **90** may be disposed above the tub **20** and supply the detergent into the tub **20** during the water supplying process.

(43) In the clothes treating apparatus according to an embodiment of the present disclosure 1, water flowing into the washing machine **1** through the water supply device **14** may flow into the detergent supply devices **90** and **100**. For example, the second detergent supply device **90** may be connected to the tub **20** via a supply pipe **17**. Wash water supplied via the water supply pipe **14a** may pass through the second detergent supply device **90** and be mixed with the detergent, and the mixed water in which the wash water is mixed with the detergent may be supplied into the tub **20**.

(44) The washing machine may further include a circulation device **200**. The circulation device **200** may allow water contained in the tub **20** to circulate inside the washing machine **1**. For example, the circulation device **200** may allow water to flow from the tub **20** into the first detergent supply device **100** and then to flow from the first detergent supply device **100** back into the tub **20**.

(45) The drive device **60** may be disposed behind the drum **30**. The drive device **60**, as a component to rotate the drum **30**, may be configured to rotate the drum **30** by transmitting a driving force generated by a motor **61** to a rotary shaft **67**.

(46) The motor **61** including a fixed stator **63** and a rotor **65** configured to rotate via electromagnetic interaction with the stator **63** may convert electric power into mechanical rotational power. However, the components of the motor **61** are not limited thereto.

(47) The tub **20** stores the mixed water in which wash water is mixed with the detergent and may have a substantially cylindrical shape. The tub **20** may be fixed inside the cabinet **10**. The opening **13** of the front panel **11** may be connected to the tub **20** by a diaphragm **50**. The diaphragm **50** may seal a space between the front panel **11** and the tub **20**.

(48) A spray nozzle **70** connected to a circulation hose and configured to spray the circulating mixed water into the tub **20** and/or the drum **30** may be provided at one side of the diaphragm **50**. The spray nozzle **70** may be provided to uniformly spray the mixed water into the entire laundry contained in the drum **30**. The circulation hose may be connected to a circulation pump **210** configured to pump the contained mixed water from the bottom of the tub **20**.

(49) The drum **30** may perform a washing operation by lifting and dropping laundry while rotating in the tub **20**. A plurality of lifters **31** may be provided inside the drum **30**.

(50) A drain device (not shown) configured to drain water contained in the tub **20** and including a drain pipe **241** (See FIG. 6) and a drain valve (not shown) may be installed below the tub **20**.

(51) The tub **20** may be provided to be elastically supported by the cabinet **10** via a spring (not shown) located at an upper portion of the tub **20** and dampers **80** provided at lower portions of the tub **20**. That is, the spring and the dampers **80** absorb vibration energy between the tub **20** and the cabinet **10** while vibration generated during rotation of the drum **30** is transmitted to the tub **20** and the cabinet **10**, thereby attenuating the vibration transmitted to the tub **20** and the cabinet **10**.

(52) The washing machine **1** may include a hot air supply device **170**. The hot air supply device

170 may be disposed in the cabinet **10**. The hot air supply device **140** may supply hot air to the tub **20** and/or the drum **30**. The hot air supply device **170** may include a fan **172** and a heat exchanger **171**.

(53) The washing machine **1** according to an embodiment of the present disclosure may include a compressor (not shown), a heat exchanger **171**, and an expansion device (not shown). The heat exchanger **171** may be located above the drum **30**. The heat exchanger **171** may include a condenser (not shown) and an evaporator (not shown).

(54) The washing machine may further include the fan **172**. The fan **172** may blow air inside the drum interior **30a** and air passing through the heat exchanger **171** in the air circulation path **190**. The fan **172** may form an air flow such that air flows in the air circulation path **190**. The fan **172** may be a centrifugal fan. However, types of the fan **172** are not limited thereto.

(55) The heat exchanger **171** may be configured to dehumidify and heat air passing through the air circulation path **190**. The heat exchanger **171** may dehumidify and heat air flowing into a tub interior **20c**. Thereby, clothes contained in the drum interior **30a** may be dried.

(56) The washing machine may include a duct assembly **180**. The duct assembly **180** may further include an upper duct **181**, a front duct **182**, and a rear duct **183**.

(57) The upper duct **181** may be located above the drum **30** and the tub **20** and guide air discharged from the drum **30** to flow back into the tub **20** after dehumidifying and heating the air. The heat exchanger **171** may be accommodated in the upper duct **181**. The upper duct **181** may be referred to as an upper frame **181**. A first air circulation path **191** may be provided in the upper duct **181**.

(58) The front duct **182** may be located in front of the drum **30** and the tub **20** and guide air flowing toward the drum **30**. Air that has passed the heat exchanger **171** may be supplied into the drum **30** through the front duct **182**. The front duct **182** may constitute a part of the air circulation path **190**. The fan **172** may be accommodated in the front duct **182**. A second air circulation path **192** may be provided in the front duct **182**.

(59) The rear duct **183** may be located behind the drum **30** and the tub **20** and guide air from the drum interior **30a** to the heat exchanger **171**. Air contained in the drum interior **30a** may flow into the heat exchanger **171** through the rear duct **183**. The rear duct **183** may constitute a part of the air circulation path **190**. A third air circulation path **193** may be provided in the rear duct **183**.

(60) The front duct **182** and the rear duct **183** may guide air contained in the drum interior **30a** to circulate through the air circulation path **190** in the cabinet **10**. In addition, because air discharged from the front duct **182** may be introduced into the tub **20** and the drum **30** through the diaphragm **50**, the diaphragm **50** may also circulate air inside the cabinet **10** through the air circulation path **190**.

(61) The washing machine may further include the air circulation path **190**. The air circulation path **190** may include a first air circulation path **191**, a second air circulation path **192**, and a third air circulation path **193**. The first air circulation path **191** may be formed by the upper duct **181**, the second air circulation path **192** may be formed by the front duct **182**, and the third air circulation path **193** may be formed by the rear duct **183**. The first air circulation path **191**, the second air circulation path **192**, and the third air circulation path **193** are not limited to the above-described example. For example, a flow path formed in the rear duct **183** may be the first air circulation path **191**, a flow path formed in the upper duct **181** may be the second air circulation path **192**, and a flow path formed in the front duct **182** may be formed in the third air circulation path **193**.

(62) Hereinafter, an air flow in the washing machine while the drying mode of the washing machine is performed will be described. After the washing and spin-drying modes of the washing machine are finished, a drying mode may be performed. A user may add the drying mode before the washing mode of the washing machine is initiated.

(63) Air in the drum interior **30a** may circulate in a closed circuit (in a direction of the arrow) inside the cabinet **10**. In response to the operation of the fan **172**, humid air in the drum interior **30a** may flow into the tub **20** through spin-drying holes **32** formed in the circumferential inner wall of

the drum **30** and/or through-holes **33** formed in a drum rear wall **30b**. For example, air in the drum interior **30a** may flow into a space **20a** between the drum **30** and the tub **20**. Air in the space **20a** between the drum **30** and the tub **20** may flow into the air circulation path **190** through communicating holes **183c** formed in a tub rear wall **20b**. Air may pass through the third air circulation path **193** located behind the tub **20** and the drum **30**. After air flows to the top end of the third air circulation path **193**, the air may flow in the first air circulation path **191**. The air may sequentially pass through the heat exchanger **171** in the first air circulation path **191**. Air having a high humidity by the clothes of the drum interior **30a** may become hot air having a low humidity while passing through the heat exchanger **171**. The hot air with a low humidity may be blown by the fan **172** disposed at a downstream area of the heat exchanger **171**. Air that has passed through the fan **172** may flow in the second air circulation path **192** in front of the tub **20** and the drum **30**. Air in the second air circulation path **192** may pass through a communicating hole **182c** formed at the diaphragm **50** at the bottom end of the second air circulation path **192** and flow back into the drum **30** after passing through the diaphragm **50**. The hot air with a low humidity may become humid air again by the clothes contained in the drum interior **30a** and then turn to hot air with a low humidity again by the above-described process.

(64) FIG. 5 is a perspective view showing a state where the cabinet is omitted from the clothes treating apparatus illustrated in FIG. 1. FIG. 6 is a perspective view of the clothes treating apparatus illustrated in FIG. 5. FIG. 7 is a block diagram schematically illustrating the clothes treating apparatus illustrated in FIG. 1.

(65) Referring to FIGS. 5 to 7, the clothes treating apparatus according to an embodiment of the present disclosure may include a water supply device **14**, a tub **20**, a circulation device **200**, and a detergent supply device **100**. In the following description, the detergent supply device **100** may be the first detergent supply device **100**.

(66) The circulation device **200** may include a circulation pump **210**, a first circulation path **220**, and a second circulation path **230**. The circulation device **200** may allow water contained in the tub interior **20c** to flow out of the tub **20** and back to the tub interior **20c**. For example, the circulation device **200** may be disposed below the tub **20** and circulate water contained in the tub interior **20c**. However, the position of the circulation device **200** is not limited to the above-described example.

(67) In addition, although a plurality of separate circulation paths **220** and **230** are illustrated in the drawings, the embodiment is not limited thereto, and the circulation paths **220** and **230** may be integrated with each other. For example, the first circulation path **220** or the second circulation path **230** may be omitted and wash water may circulate via another circulation path. The plurality of circulation paths **220** and **230** may be referred to as a plurality of circulation guides **220** and

(68) The circulation pump **210** may be connected to the plurality of circulation paths **220** and **230** to allow water contained in the circulation paths **220** and **230** to flow into tub **20**. The circulation pump **210** may be located at one side of the plurality of circulation paths **220** and **230**. For example, the circulation pump **210** may be located on the left side (e.g., in the Y direction) of the first circulation path **220** under the second circulation path **230**. The circulation pump **210** may operate optionally with a drain pump **240**, which will be described below.

(69) The first circulation path **220** may be provided to guide wash water from the tub **20** to the circulation pump **210**. The first circulation path **220** may be disposed under the tub **20** and guide water drained from the tub interior **20c** to the circulation pump **210**. The first circulation path **220** may include a drain path **221** and a mixing path **222**. The first circulation path **220** may be referred to as a first circulation guide **220**.

(70) The drain path **221** may be disposed under the tub **20** and guide water drained from the tub interior **20c**. An inner flow path **224** may be formed in the drain path **221**. The drain path **221** may guide water drained from the tub **20** to flow into the circulation pump **210** or to be discharged out of the washing machine **1**. The drain path **221** may be referred to as a drain guide **221**.

(71) The drain path **221** may include a first drain path **221a** and a second drain path **221b**. The first

drain path **221a** may guide water discharged from the tub **20** to the second drain path **221b**. The second drain path **221b** may guide water discharged from the tub **20** to flow into the circulation pump **210** or to be discharged out of the washing machine. The inside of the second drain path **221b** may be open or closed by the drain pump **240**. The second drain path **221b** may serve as a filter case **234** to which a filter is coupled. The drain pump **240** may be coupled to the second drain path **221b**. The first drain path **221a** may be referred to as a first drain guide **221a** and the second drain path **221b** may be referred to as a second drain guide **221b**.

(72) The mixing path **222** may be connected to the drain path **221**. For example, the mixing path **222** may be connected to the second drain path **221b** and guide wash water to the circulation pump **210**. The mixing path **222** may be connected to the detergent supply device **100**. For example, a detergent contained in the detergent case **110** of the detergent supply device **100** may be sucked into a detergent pump **120** and discharged into a mixing space **222a** in the mixing path **222**.

Detailed descriptions thereof will be given below. Therefore, wash water and the detergent may be mixed in the mixing path **222**, and the wash water mixed with the detergent may flow into the tub **20** after passing through the mixing path **222**, the circulation pump **210**, and the second circulation path **230**. The mixing path **222** may be mounted on the detergent pump **120** which will be described below. Thus, the mixing path **222** may be directly connected to the detergent pump **120** without a separate part connecting the mixing path **222** with the detergent pump **120**, and detergent residues may be minimized. The mixing path **222** may be referred to as a mixing guide **222**.

(73) The second circulation path **230** may be provided to guide wash water from the circulation pump **210** to the tub **20**. The second circulation path **230** may be disposed under the tub **20** and guide wash water from the circulation device **200** to the tub interior **20c**. The second circulation path **230** may be connected to the first circulation path **220** and the tub **20**, respectively. The second circulation path **230** may be referred to as a second circulation guide **230**. The second circulation path **230** may include a circulation pump case **231** and a supply pipe **232**. Detailed descriptions thereof will be given below.

(74) The detergent supply device **100** may be disposed at one side of the circulation device **200**. The detergent supply device **100** may supply the detergent to the circulation device **200**. The detergent supplied to the circulation device **200** may be mixed with wash water in the circulation device **200**. The mixed detergent may flow into the tub **20** through the plurality of circulation paths **220** and **230**.

(75) The detergent supply device **100** may include the detergent pump **120** configured to suck a detergent from the detergent case **110** after docking with the detergent case **110**, a housing **160** accommodating the detergent case **110** and provided to be withdrawn from the cabinet **10** and inserted into the cabinet **10**, and a second drawer **150** accommodating the housing **160**. The housing **160** may slide out from the inside of the second drawer **150**. The second drawer **150** may be coupled to side walls **12b** and **12c** or side plates **12b** and **12c** of the cabinet **10**.

(76) The drain pump **240** may drain water discharged from the tub **20** out of the washing machine through the drain pipe **241**. The drain pump **240** may be coupled to the second drain path **221b** to open or close the inner flow path **224** in the second drain path **221b**. The drain pump **240** may be turned on/off optionally with the circulation pump **210** by the detergent pump **120**. For example, to circulate water in the washing machine, the circulation pump **210** may be driven. On the contrary, to drain water, only the drain pump **240** may be driven, and therefore water may be drained out of the washing machine.

(77) The washing machine may further include a switch. The switch may sense whether a first drawer is accommodated in the cabinet **10**. For example, in response to insertion of that the second drawer **150** into the cabinet **10**, a button (not shown) of the switch may be in a pressed state. In response to withdrawing of the second drawer **150** from the cabinet **10**, the button (not shown) may be in a non-pressed state.

(78) The washing machine may further include a sensor guide **225**. The sensor guide **225** may

protrude from the first drain guide **221a**. The sensor guide **225** may be connected to a water level sensor **250**. For example, the sensor guide **225** may be connected to the water level sensor **250** via a separate hose. For example, the water level sensor **250** may be an air sensor, and the water level in the sensor guide **225** may increase as the water level in the tub **20c** increases. Accordingly, a pressure in the sensor guide **225** may increase in accordance with the increasing water level, and the water level sensor **250** may sense the water level of the tub interior **20c** by sensing the same.

(79) Referring to FIGS. 5 to 7, wash water circulating in the cabinet **10** will be described.

(80) Water introduced from the outside of the washing machine **1** may enter the tub **20** through the water supply device **14**. For example, the water supply device **14** may be disposed above the tub **20** to allow water to flow into the interior **20c** of the tub **20**.

(81) The water level sensor **250** provided at one side of the tub **20** may sense an amount of water collected in the tub interior **20c**. For example, the water level sensor **250** may be coupled to an upper portion of the tub **20** and sense the amount of wash water of the tub interior **20c**. For example, upon determination that the water level of the tub **20** sensed by the water level sensor **250** is $\frac{1}{3}$ or more, a processor **300** may operate the circulation pump **210** in response thereto.

Alternatively, the processor **300** may operate the drain pump **240** to drain water contained in the tub interior **20c** out of the washing machine. In this case, the first circulation path **220**, the second circulation path **230**, and the like below the tub **20** may be filled with water in the cabinet **10**.

(82) Although the water level sensor **250** disposed in the interior **20c** of the tub **20** is illustrated in the drawing, the position of the water level sensor **250** is not limited to the above-described example.

(83) Hereinafter, the embodiment will be described based on the operation of the circulation pump **210**.

(84) In response to the operation of the circulation pump **210**, water contained in the tub interior **20c** may flow to the circulation device **200** through the drain hole **223** disposed under the tub **20**. For example, water contained in the tub interior **20c** may flow to the inner flow path **224** through the first circulation path **220** or the drain hole **223** of the drain path **221**.

(85) Wash water may be discharged from the tub **20** through the first drain path **221a** and guided to the second drain path **221b**. Because a filter (not shown) is provided in the second drain path **221b**, wash water may be filtered by the filter. Wash water guided to the second drain path **221b** may be guided to the mixing path **222**. In this regard, in response to the operation of the detergent pump **120**, the detergent may be discharged to mixing space **222a** in the mixing path **222** from the detergent pump **120**. Wash water may be mixed with the detergent while passing through the mixing path **222**.

(86) Wash water mixed with the detergent may be guided toward the circulation pump **210**. Wash water may flow to the second circulation path **230** through the circulation pump **210**. Because the second circulation path **230** may be connected to the first circulation path **220** and the tub **20**, respectively, wash water that has passed through the first circulation path **220** and the circulation pump **210** may flow into the tub interior **20c**.

(87) The embodiment will be described based on the operation of the drain pump **240**.

(88) In response to the operation of the drain pump **240**, water contained in the tub interior **20c** may flow to the circulation device **200** through the drain hole **223** disposed under the tub **20**. For example, water contained in the tub interior **20c** may flow to the inner flow path **224** through the first circulation path **220** or the drain hole **223** of the drain path **221**.

(89) Wash water may be discharged from the tub **20** through the first drain path **221a** and guided to the second drain path **221b**. Because a filter (not shown) is provided in the second drain path **221b**, wash water may be filtered by the filter. The drain pump **240** may serve as a valve to open and close the inner flow path **224**, and the inner flow path **224** may be open and closed by operating the drain pump **240**. In response to the opening of the inner flow path **224**, wash water guided to the second drain path **221b** may be guided out of the washing machine **1** through the drain pipe **241**. In

this case, the detergent pump **120** and/or the circulation pump **210** may be in an off-state.

(90) FIG. **8** is a perspective view of the detergent supply device and the circulation device of the clothes treating apparatus illustrated in FIG. **1**. FIG. **9** is an exploded perspective view of the detergent supply device and the circulation device illustrated in FIG. **8**.

(91) Referring to FIGS. **8** and **9**, the clothes treating apparatus according to an embodiment of the present disclosure may include the detergent supply device **100** and the circulation device **200**.

(92) The detergent supply device **100** may include a detergent case **110**, a detergent pump **120**, a detergent case cover **130**, a cap **140**, a second drawer **150**, and a housing **160**.

(93) The detergent case **110** may be docked with the detergent pump **120**. Upon docking of the detergent case **110** with the detergent pump **120**, the detergent of the accommodating space **110a** of the detergent case **110** may be discharged into the mixing path **222** via the detergent pump **120**.

(94) The detergent case **110** may be inserted into the second drawer **150** or withdrawn from the second drawer **150** in a state of being accommodated in the housing **160**. The detergent case **110** may be inserted into a receiving space **151** or withdrawn from the receiving space **151** via the opening **150a** of the second drawer **150**. The detergent case **110** and the housing **160** may be slid out from the second drawer **150**.

(95) The detergent case **110** may include a first detergent case **111**, a second detergent case **112**, and a pump connector **111b**.

(96) The detergent case **110** may store a main detergent (powder detergent or liquid detergent), a preliminary detergent, a fabric softener and/or a bleach.

(97) A main laundry detergent, a fabric softener, and/or a rinse may be contained in the first detergent case **111** and/or the second detergent case **112**. For example, the main laundry detergent may be contained in the first detergent case **111**, and the fabric softener or the preliminary detergent may be contained in the second detergent case **112**. However, types of the detergent contained therein and/or size of the detergent case **110** are not limited to the above-described example. The first detergent case **111** and the second detergent case **112** may be disposed inside the housing **160**. The first detergent case **111** may form a first accommodating space **111a** and the second detergent case **112** may form a second accommodating space **112a**. For example, the main laundry detergent may be contained in the first accommodating space **111a**, and the fabric softener, the preliminary detergent, or the like may be contained in the second accommodating space **112a**.

(98) Although the drawings illustrate a plurality of detergent cases **110** partitioned and separated from each other, the embodiment is not limited thereto.

(99) The pump connector **111b** may be formed at the rear of the first detergent case **111** and at the rear of the second detergent case **112**, respectively. The pump connector **111b** may serve as an outlet **111b**. Although the outlet **111b** is formed only at the first detergent case **111** in the drawing, the outlet **111b** may also be formed at the second detergent case **112**. The outlet **111b** may allow the detergent contained in the first accommodating space **111a** and/or the second accommodating space **112a** to flow into the detergent pump **120** via a pipe (not shown). For example, upon docking of the detergent case **110** with the detergent pump **120**, the detergent may flow into the detergent pump **120** through the outlet **111b** and then into the mixing space **222a** of the mixing path **222**. However, the position of the outlet **111b** is not limited to that shown in the drawing.

(100) The detergent case cover **130** may be configured to cover the top of the detergent case **110**. For example, the detergent case cover **130** may be configured to cover the top end of the detergent case **110**. The detergent case cover **130** may include a first detergent case cover **131** configured to cover the top of the first detergent case **111** to correspond to the first accommodating space **111a**, and a second detergent case cover **132** configured to cover the top of the second detergent case **112** to correspond to the second accommodating space **112a**.

(101) The cap **140** may be coupled to the detergent case cover **130**. The cap **140** may be removed in the case where the user adds a detergent into the interior **111a** and **112a** of the detergent case **110**. Upon completion of adding the detergent into the interior **111a** and **112a** of the detergent case **110**

by the user, the cap **140** may be mounted on the detergent case cover **130** again. The interior **111a** and **112a** of the detergent case **110** may serve as the accommodating spaces **111a** and **112a** of the detergent case **110**.

(102) The cap **140** may include a first cap **141** and a second cap **142**. The first cap **141** may be mounted on the first detergent case cover **131**, and the second cap **142** may be mounted on the second detergent case cover **132**.

(103) The detergent pump **120** may be disposed behind the second drawer **150** and the detergent case **110**. Upon docking with the detergent case **110** inserted into the second drawer **150**, the detergent pump **120** may suck the detergent contained in the detergent case **110** and discharge the sucked detergent into the mixing space **222a** in the mixing path **222**. For example, upon docking of the outlet **111b** of the detergent case **110** with first mover **121a** of the detergent pump **120**, the detergent contained in the detergent case **110** may flow into the detergent pump **120** through the outlet **111b** and the first mover **121a** and then flow into the mixing space **222a** through the second mover **121b**. The detergent flowing into the mixing space **222a** may be mixed with water entering the tub interior **20c** through the mixing path **222**.

(104) The second drawer **150** may accommodate the housing **160** and the detergent case **110**. The housing **160** and the detergent case **110** may be withdrawn from or inserted into the second drawer **150** via the opening **150a**. The second drawer **150** may serve as the case **150**. The second drawer **150** is coupled to a side wall of the cabinet **10** to accommodate the first drawer and may be slidably coupled to the first drawer such that the first drawer slides.

(105) The second drawer **150** may include a receiving space **151**, a sliding coupler **152**, a filter coupler **153**, a rail coupler **154a**, and a stopper **155**.

(106) The receiving space **151** may be formed inside the second drawer **150**. The housing **160** and the detergent case **110** may be disposed in the receiving space **151**.

(107) The sliding couplers **152** may be formed from both side walls **154** of the second drawer **150**. The sliding couplers **152** may extend backward from both side walls **154**. The sliding couplers **152** may be coupled to side plates **12b** and **12c** of the cabinet **10**. The sliding coupler **152** may be provided in plural. A plurality of sliding couplers **152** may extend backward from the left wall **154** and/or the right wall **154** of the second drawer **150**.

(108) The filter coupler **153** may be formed by opening a part of a rear wall **150b** of the second drawer **150**. The filter case **234** may be coupled to the filter coupler **153**.

(109) The rail coupler **154a** may be formed by being recessed from an inner wall **154** of the second drawer **150**. The rail coupler **154a** may be provided in plural. For example, the rail coupler **154a** may be formed by being recessed from inner surfaces of the left wall **154** and/or the right wall **154** of the second drawer **150**. The rail coupler **154a** may be slidably coupled to a rail **164** of the housing **160**. For example, the rail **164** of the housing **160** may be coupled to the rail coupler **154a** and the housing **160** may slide.

(110) The stopper **155** may fix the rail **164** in a state where the second drawer **160** is inserted into or accommodated in the cabinet **10**. The stopper **155** may be mounted at the sliding coupler **152**. The stopper **155** may be provided in plural.

(111) The housing **160** may accommodate the detergent case **110**, the detergent case cover **130**, and the cap **140**. The housing **160** may be slidably coupled to the second drawer **150**. The user may add the detergent into the interior **111a** and **112a** of the detergent case **110** accommodated in the housing **160** by pulling the housing **160** from the cabinet **10** and then insert the housing **160** into the cabinet **10** again.

(112) The housing **160** may serve as a first drawer. The housing **160** may be provided to be withdrawn from or inserted into the cabinet **10**.

(113) The housing **160** may be provided in plural. The plurality of housings **160** may include a first housing **161** and a second housing **162**. The first housing **161** may accommodate the first detergent case **111**, and the second housing **162** may accommodate the second detergent case **112**. Each of the

housings **161** and **162** may include a handle **165a** or **165b**, an accommodating space **161c** and **162c**, and a rail **164**.

(114) The handles **165a** and **165b** may be formed at the front surfaces **161a** and **161b** of the housing **160**. The handles **165a** and **165b** may be recessed from the front surfaces **161a** and **161b**. The handles **165a** and **165b** may be provided in plural. The first handle **165a** may be formed at the front surface **161a** of the first housing **161**, and the second handle **165b** may be formed at the front surface **161b** of the second housing **162**. The user may hold the detergent case **110** via the handles **165a** and **165b** and insert the detergent case **110** into the cabinet **10** or withdraw the detergent case **110** from the cabinet **10**.

(115) The second housing **162** may accommodate the detergent case **110**, the detergent case cover **130**, and the cap **140**. The top end of the second housing **162** may be open. The detergent case **110**, the detergent case cover **130**, and the cap **140** may be separated from the housing **160** through the top side of the accommodating space **162**.

(116) The rail **164** may be coupled to a side wall **163** of the housing **160**. The rail **164** may be provided in plural. For example, the first rail **164** may be coupled to a left wall **163** of the first housing **161**, and the second rail **164** may be coupled to a right wall of the second housing **162**.

(117) The circulation device **200** may circulate water contained in the tub interior **20c**. For example, the circulation device **200** may allow water contained in the tub interior **20c** to flow out of the tub **20** and back into the tub interior **20c**. The circulation device **200** may include the circulation pump **210**, the first circulation path **220**, and the second circulation path **230**.

(118) The circulation pump **210** may be disposed on the left below the tub **20**. The circulation pump **210** may pump wash water to introduce wash water flowing in the circulation device **200** into the tub **20**. The circulation pump **210** may be mounted on the circulation pump case **231**. For example, the circulation pump **210** may be mounted on a circulation pump mount **231a** provided at the circulation pump case **231**. The circulation pump **210** may open or close flow paths **222a** and **232a** provided in the circulation device **200**.

(119) The first circulation path **220** may allow wash water to flow into the circulation pump **210** from the tub **20**. The first circulation path **220** may include the drain path **221** and the mixing path **222**.

(120) The drain path **221** may allow water discharged from the tub **20** to flow into the circulation pump **210** or to flow out of the washing machine. The drain path **221** may include the first drain path **221a** and the second drain path **221b**.

(121) The first drain path **221a** may guide water discharged from the tub **20** to the second drain path **221b**. A first end of the first drain path **221a** may be connected to the tub **20**, and a second end **226** of the first drain path **221a** may be connected to the second drain path **221b**. The drain hole **223** may be formed at the first end of the first drain path **221a**. The inner flow path **224** may be formed in the first drain path **221a**.

(122) The second drain path **221b** may guide water discharged from the tub **20** to flow into the circulation pump **210** or to flow out of the washing machine. The second drain path **221b** may be connected to the first drain path **221a**. The second drain path **221b** may branch the inner flow path **224** formed in the first drain path **221a**. The second drain path **221b** may serve as a filter case **234** to which a filter is coupled.

(123) The filter case **234** may include a filter mount **234a**, a drain pump mount **234b**, a mixing guide mount **234c**, a first drain guide connector **234d**, and a spray nozzle connector **234e**. Also, the drain pipe **241** (See FIG. 6) may be formed at one side of the filter case **234**. The drain pipe **241** may be a first drain pipe **241**.

(124) The filter mount **234a** may be formed in the filter case **234**. A filter (not shown) may be mounted on the filter mount **234a** to filter water discharged from the tub **20**.

(125) The drain pump **240** may be mounted on the drain pump mount **234b**. The drain pump **240** may open or close the inner flow path **224**. For example, the drain pump **240** may open or close the

flow path in the filter case **234**. In response to the operation of the drain pump **240**, water that has passed through the first drain path **221a** and the second drain path **221b** may flow out of the washing machine. The drain pump mount **234b** may be formed on the right side of the filter case **234**.

(126) The mixing path **222** may be mounted on the mixing guide mount **234c**. For example, the mixing guide mount **234c** may be connected to a second drain guide connector **222b** of the mixing path **222**. Accordingly, water that has passed through the drain path **221** may flow into the mixing path **222**. The mixing guide mount **234c** may be formed on the left side of the filter case **234**.

(127) The first drain guide connector **234d** may be connected to the first drain path **221a**. For example, the first drain guide connector **234d** may be connected to the second end **226** of the first drain path **221a**.

(128) The spray nozzle connector **234e** may be connected to the spray nozzle **70** spraying wash water into the tub **20** and/or the drum **30**. The spray nozzle connector **234e** may protrude upward.

(129) The mixing path **222** may guide water that has passed through the drain path **221** to the circulation pump **210**. The mixing space **222a** may be formed in the mixing path **222**. In the mixing space **222a**, wash water may be mixed with the detergent introduced into the mixing space **222a** from the detergent case **110** through the detergent pump **120**. The mixing path **222** may be disposed under the detergent case **110** and the detergent pump **120** (See FIGS. **11** and **12**). The mixing path **222** may be directly connected to the detergent pump **120** without using a hose or pipe interposed therebetween. Therefore, the possibility of solidification of a detergent remaining in a hose or pipe, which connects the mixing path **222** with the detergent pump **120**, may be significantly reduced.

(130) The mixing path **222** may include a mixing space **222a**, a second drain guide connector **222b**, a circulation pump communicator **222c**, and a drain pipe **222e**. The mixing space **222a** may be formed in the mixing path **222** and wash water may be mixed with the detergent therein.

(131) The second drain guide connector **222b** may be disposed on the right side of the mixing path **222** and connected to the mixing guide mount **234c**. Thereby, wash water contained in the drain path **221** may flow into the mixing space **222a** and the detergent may be mixed with wash water.

(132) The circulation pump communicator **222c** may be disposed on the left side of the mixing path **222** to be connected to the circulation pump **210** and/or the circulation pump case **231**. Thereby, wash water contained in the mixing space **222a** may be pumped by the circulation pump **210** to flow into the tub **20**.

(133) The drain pipe **222e** may be a second drain pipe **222e**. The second drain pipe **222e** may be provided to remove water remaining after the rinsing operation of the washing machine. For example, the second drain pipe **222e** may be blocked with a plug, and the user may remove water remaining in the mixing space **222a** by removing the plug, if required.

(134) The second circulation path **230** may be provided to guide wash water from the circulation pump **210** to the tub **20**. The second circulation path **230** may be connected to the first circulation path **220** and the tub **20**, respectively. The second circulation path **230** may include the circulation pump case **231** and the supply pipe **232**.

(135) The circulation pump case **231** may accommodate a part of the circulation pump **210**. The circulation pump case **231** may include a circulation pump mount **231a** and a supply pipe mount **231b**.

(136) The circulation pump mount **231a** may be formed at one side of the circulation pump case **231** in an open state. For example, the circulation pump mount **231a** may be formed on the left side of the circulation pump case **231**. The circulation pump **210** may be mounted on the circulation pump mount **231a**.

(137) The supply pipe mount **231b** may extend upward from the circulation pump case **231**. The supply pipe mount **231b** may protrude toward the supply pipe **232**. For example, the supply pipe mount **231b** may be connected to a first end **232b** of the supply pipe **232**. The first end **232b** of the supply pipe **232** may be a lower end **232b**. The lower end **232b** of the supply pipe **232** may serve as

a circulation pump connector **232b**.

(138) The supply pipe **232** may allow wash water pumped by the circulation pump **210** to flow into the tub **20** from the circulation pump case **231**. A supply flow path **232a** may be formed in the supply pipe **232**. The supply flow path **232a** may connect the mixing space **222a** with the interior **20c** of the tub **20**. The supply pipe **232** may be connected to the circulation pump **210** and the tub **20**, respectively. The first end **232b** of the supply pipe **232** may serve as a circulation pump connector **232b** connected to the circulation pump **210**. A second end **232c** of the supply pipe **232** may serve as a tub connector **232c** connected to the tub **20**. The second end **232c** of the supply pipe **232** may be an upper end **232c**.

(139) FIG. **10** is an exploded perspective view of the detergent pump illustrated in FIG. **9**.

(140) Referring to FIG. **10**, the detergent supply device **100** may include the detergent pump **120** in the washing machine according to an embodiment of the present disclosure.

(141) The detergent pump **120** may include a pump **120a** and a mounting member **123**. The mounting member **123** may be mounted on the front of the pump **120a**. The mounting member **123** may be a cover member **123** covering the pump **120a**.

(142) A mover **121** may be formed at the front of the pump **120a**. The mover **121** may include a first mover **121a** and a second mover **121b**. The first mover **121a** may allow the detergent to flow into the pump **120a** from the detergent case **110**. The second mover **121b** may allow the detergent to flow into the mixing space **222a** from the pump **120a**. The first mover **121a** and the second mover **121b** may protrude forward. However, the shapes of the first mover **121a** and the second mover **121b** are not limited to the above-described example.

(143) The mounting member **123** may support the pump **120a**. In addition, the mounting member **123** may cover the front of the pump **120a** to protect the second mover **121b** and allow the second mover **121b** to communicate with the mixing path **222**. The mounting member **123** may be mounted on the mixing path **222**.

(144) The mounting member **123** may include supports **123c** and **123d** and a mount. The supports **123c** and **123d** may be provided in plural. The plurality of supports **123c** and **123d** may include a first support **123c** and a second support **123d**.

(145) The first support **123c** may support the pump **120a** at the bottom plate **19** and/or bottom. The first support **123c** may vertically extend. The second support **123d** may be bent from the top end of the first support **123c** and extend backward. The second support **123d** may support the bottom of the pump **120a**.

(146) The mounts **123e** and **123f** may be provided in plural. The plurality of mounts **123e** and **123f** may include a first mount **123e** and a second mount **123f**.

(147) The first mount **123e** may allow the second mover **121b** to communicate with the mixing path **222**. The first mount **123e** may form an outlet **123a**. The detergent contained in the detergent pump **120** may flow into the mixing path **222** through the outlet **123a**. The first mount **123e** may protrude forward from the second mount **123f**. However, the shape of the first mount **123e** is not limited to the example shown in the drawing. The first mount **123e** may allow the mounting member **123** to be mounted on the mixing path **222**. The first mount **123e** may protect the second mover **121b** and cover the second mover **121b**. The first mount **123e** may serve as a cover **123e**.

(148) The second mount **123f** may be formed near the first mount **123e**. The second mount **123f** may allow the mounting member **123** to be mounted on the pump **120a**. Ends of the second mount **123f** may extend toward the pump **120a** to be mounted on the pump **120a**.

(149) FIG. **11** is a cross-sectional view of the detergent supply device and the circulation device illustrated in FIG. **8**. FIG. **12** is a cross-sectional view of the detergent supply device and the circulation device illustrated in FIG. **8**. FIG. **11** is a cross-section view of the detergent supply device and the circulation device taken along line A-A' of FIG. **8**. FIG. **12** is a cross-sectional view of the detergent supply device and the circulation device taken along line B-B' of FIG. **8**.

(150) Referring to FIGS. **11** and **12**, the detergent supply device **100** according to an embodiment

of the present disclosure may include a detergent case **110**, a barrier wall **166** partitioning a plurality of detergent cases **110**, an accommodating space **110a** formed in the detergent case **110**, a detergent pump **120** configured to suck a detergent contained in the accommodating space **110a** and discharge the detergent into a mixing space **222a** in response to docking with the detergent case **110**, a housing **160** accommodating the detergent case **110** and allowing the detergent case **110** to slide in a forward/backward direction, and a case (second drawer) accommodating the detergent case **110** and the housing **160**.

(151) In addition, the detergent pump **120** may include a pump **120a**, and a mounting member **123** coupled to the pump **120a** in front of the pump **120a** and mounted on the mixing path **222**. The mounting member **123** may be located on the mixing path **222**. Therefore, the detergent passing through the interior **123b** of the mounting member **123** may immediately flow into the mixing space **222a** where wash water flows, and thus an amount of the detergent remaining between the detergent pump **120** and the mixing path **222** may be minimized.

(152) The detergent pump **120** may include a connector **127** electrically and/or physically connected to a device outside the detergent pump **120**. The connector **127** may be disposed behind the pump **120a**.

(153) In addition, the circulation device **200** according to an embodiment of the present disclosure may include the mixing path **222** constituting the mixing space **222a**.

(154) Referring to FIGS. **11** and **12**, a flow path of the detergent contained in the accommodating space **110a** of the detergent case **110** will be described.

(155) The accommodating space **110a** may store the detergent. In response to docking of the detergent case **110** with the detergent pump **120** or the operation of the detergent pump **120**, the detergent may be introduced into the pump **120a** through the mover **121**. For example, the detergent may be introduced into the flow path **121c** via the inlet **121d** and enter the pump **120a** by pushing a first valve **125**. The detergent introduced into the pump **120a** may be discharged out of the pump **120a** by pushing a second valve **126**. The detergent discharged from the pump **120a** may flow into the mixing space **222a** via the outlet **123a** formed at one end of the mounting member **123** from the interior **123b** of the mounting member **123**. The outlet **123a** may communicate with the mixing space **222a**. For example, the outlet **123a** may be connected to the inlet **222d**.

(156) Because wash water circulating in the circulation device **200** pass through the mixing space **222a**, the detergent may be immediately mixed with wash water. Wash water mixed with the detergent, i.e., mixed water, may flow into the tub interior **20c** through the circulation pump **210** and the second circulation path **230**.

(157) FIG. **13** is a control block diagram of the clothes treating apparatus illustrated in FIG. **1**. FIG. **14** is a control flow chart of the clothes treating apparatus illustrated in FIG. **1**.

(158) Referring to FIGS. **13** and **14**, the washing machine according to an embodiment of the present disclosure may include a processor **300**. The processor **300** may control the circulation pump **210**, the detergent pump **120**, and the drain pump **240**. For example, the processor **300** may turn on/off the circulation pump **210**, the detergent pump **120**, and/or the drain pump **240**. In addition, the processor **300** may optionally operate the circulation pump **210** or the drain pump **240**. For example, the processor **300** may turn on the circulation pump **210** and turn off the drain pump **240**. On the contrary, the processor **300** may turn off the circulation pump **210** and turn on the drain pump **240**.

(159) Referring to FIGS. **7** to **14**, a sequence of controlling the operation of the circulation device **200** in the washing machine according to an embodiment of the present disclosure will be described.

(160) Water (wash water) may be supplied to the washing machine **1** from the outside through the water supply device **14**. The water supply device **14** may supply wash water into the interior **20c** of the tub **20** (**1410**). In this case, because flow paths connected to the tub **20** are closed, water may be collected in the tub **20**. The processor **300** may sense a water level of the tub interior **20c** using the

water level sensor **250** provided at an upper portion of the tub **20**. In the case where the water level of the tub interior **20c** reaches a predetermined water level, the circulation pump **210** may operate (**1420**). For example, upon determination that the water level of the tub interior **20c** reaches $\frac{1}{3}$ or more of a height of the tub **20**, the processor **300** may operate the circulation pump **210** (**1430**). In response to the operation of the circulation pump **210**, water contained in the tub interior **20c** may flow into the circulation device **200** disposed below the tub **20**. For example, upon the operation of the circulation pump **210**, water contained in the tub interior **20c** may be guided through the drain path **221** to flow toward the mixing path **222**.

(161) For example, wash water may be discharged from the tub **20** through the first drain path **221a** and guided to the second drain path **221b**. Wash water guided to the second drain path **221b** may be guided to the mixing path **222**.

(162) In response to the operation of the circulation pump **210**, the detergent pump **120** may operate (**1440**). The detergent pump **120** may suck the detergent from the detergent case **110** and migrate the detergent to the mixing space **222a** formed in the mixing path **222** (See FIGS. **11** and **12**). For example, the detergent may be discharged into the mixing space **222a** of the mixing path **222** from the detergent pump **120**. Accordingly, wash water passing through the mixing space **222a** may be mixed with the detergent discharged from the detergent pump **120**.

(163) Wash water mixed with the detergent may be guided to the circulation pump **210**. The wash water may flow to the second circulation path **230** after passing through the circulation pump **210**. Because the second circulation path **230** is connected to the first circulation path **220** and the tub **20**, respectively, wash water that has passed through the first circulation path **220** and the circulation pump **210** may flow to the tub interior **20c**.

(164) On the contrary, the processor **300** may also operate the drain pump **240** without operating the circulation pump **210**. As described in FIG. **7**, in response to the operation of the drain pump **240**, wash water of the tub interior **20c** may flow into the inner flow path **224** through the first circulation path **220** or the drain holes **223** of the drain path **221**.

(165) Wash water may be discharged from the tub **20** through the first drain path **221a** and guided to the second drain path **221b**. The drain pump **240** may serve as a valve opening or closing the inner flow path **224**, and the inner flow path **224** may be open or closed by operating the drain pump **240**. In response to the opening of the inner flow path **224**, wash water guided to the second drain path **221b** may be guided out of the washing machine **1** through the drain pipe **241**.

(166) Meanwhile, the washing machine may include a plurality of detergent cases **111** and **112**. In addition, the washing machine may include a plurality of modes including a washing mode, a spin-drying mode, a rinsing mode, a drying mode, and the like.

(167) In response to the operation of the circulation pump **210**, the processor **300** may operate the detergent pump **120** such that the detergent contained in the first detergent case **111** flows to the mixing space **222a** through the detergent pump **120**. Thereby, a washing mode may be performed in the tub **20** using mixed water in which the detergent and wash water are mixed. Thereafter, for example, in a final rinsing mode of the washing machine, the processor **300** may operate the detergent pump **120** such that the detergent contained in the second detergent case **112** flows to the mixing space **222a** through the detergent pump **120**. Thereby, the rinsing mode may be performed using the mixed water in which the detergent is mixed with wash water. In this case, the detergent contained in the first detergent case **111** may be a laundry detergent, and the detergent contained in the second detergent case **112** may be a fabric softener.

(168) The embodiments of the present disclosure have been shown and described above with reference to the accompanying drawings. However, the disclosed embodiments are illustrative and the scope of the present disclosure is not limited thereby. It will be understood by those of ordinary skill in the art that the present disclosure may be easily modified into other detailed forms without changing the technical principle or essential features of the present disclosure.

Claims

1. A clothes treating apparatus comprising: a cabinet; a tub in the cabinet and configured to collect wash water; a circulation device configured to circulate the wash water contained in the tub, and the circulation device comprising: a circulation pump; a first circulation path to guide the wash water from the tub to the circulation pump; and a second circulation path to guide the wash water from the circulation pump to the tub so that the wash water circulates by flowing from the tub to the circulation pump through the first circulation path and flowing back to the tub from the circulation pump through the second circulation path; and a detergent supply device configured to supply a detergent to the circulation device, the detergent supply device comprising: a detergent case to store the detergent; and a detergent pump connected to the detergent case and directly mounted on the first circulation path to supply the stored detergent from the detergent case to the first circulation path, the detergent pump including: an inlet through which the detergent is introduced from the detergent case; and an outlet through which the introduced detergent is discharged, wherein the first circulation path comprises a mixing path mounted with the detergent pump, the mixing path including an inlet which is directly connected to the outlet of the detergent pump to receive the detergent from the detergent case thereby mixing the detergent with the wash water flowing in the first circulation path.
2. The clothes treating apparatus according to claim 1, wherein the first circulation path further comprises: a drain path to guide the wash water discharged from the tub therethrough.
3. The clothes treating apparatus according to claim 2, wherein the drain path comprises: a first drain path to guide the wash water from the tub out of the tub therethrough; and a second drain path connected to the first drain path and branching from the first drain path.
4. The clothes treating apparatus according to claim 3, further comprising a drain pump configured to receive the wash water from the first drain path and mounted on the second drain path.
5. The clothes treating apparatus according to claim 4, further comprising a water supply device configured to supply the wash water into the tub from an external water supply source, wherein the wash water flowing into the tub by the water supply device flows to the tub after passing through the first circulation path, the circulation pump, and the second circulation path in response to the circulating of the wash water by the circulation pump.
6. The clothes treating apparatus according to claim 5, further comprising a processor configured to selectively operate the circulation pump or the drain pump.
7. The clothes treating apparatus according to claim 6, wherein the processor is further configured to operate the circulation pump in response to determining that a water level in the tub reaches a predetermined water level.
8. The clothes treating apparatus according to claim 7, wherein the processor is further configured to operate the detergent pump in response to the operating of the circulation pump.
9. The clothes treating apparatus according to claim 8, wherein the detergent case comprises: a first detergent case; and a second detergent case separated from the first detergent case in a lateral direction, wherein the processor is further configured: to operate the circulation pump based on an initiation of a washing operation; to operate the detergent pump to allow a detergent contained in the first detergent case to flow into the detergent pump based on the operation of the circulation pump; and to operate the detergent pump to allow a detergent contained in the second detergent case to flow into the detergent pump based on an initiation of a rinsing mode.
10. The clothes treating apparatus according to claim 6, wherein the processor is further configured to turn off the circulation pump and turn on the drain pump to drain the wash water contained in the tub.
11. The clothes treating apparatus according to claim 5, wherein the detergent supply device and the circulation device are disposed below the tub.

12. The clothes treating apparatus according to claim 11, wherein the detergent supply device is a first detergent supply device, and the clothes treating apparatus further comprises a second detergent supply device disposed above the tub to supply the detergent into the tub.
 13. The clothes treating apparatus according to claim 12, further comprising: a fan above the tub to form a circulating air flow passing through the tub; and a heat exchanger above the tub to dehumidify and heat air discharged from the tub and supply the dehumidified and heated air to the tub.
 14. The clothes treating apparatus according to claim 1, wherein the detergent supply device further comprises a drawer configured to accommodate the detergent case and to be withdrawn from or inserted into the cabinet.
 15. The clothes treating apparatus according to claim 14, wherein the drawer comprises: a first drawer configured to accommodate the detergent case and withdrawn from or inserted into the cabinet; and a second drawer coupled to a side wall of the cabinet, configured to accommodate the first drawer, and slidably coupled to the first drawer allowing the first drawer to slide.
 16. The clothes treating apparatus of claim 1, wherein the mixing path comprises a mixing space formed therein to receive the wash water from the tub and the discharged detergent through the inlet of the mixing path which is directly connected to the outlet of the detergent pump.
 17. The clothes treating apparatus of claim 16, wherein the supplied wash water from an external water supply source flows back to the tub after passing through the water supply device, the tub, and the mixing space.
 18. A clothes treating apparatus comprising: a cabinet; a tub in the cabinet and configured to collect wash water; a detergent supply device configured to supply a detergent into the tub, the detergent supply device including: a detergent case to accommodate the detergent; a detergent pump including: an inlet through which the detergent is introduced from the detergent case; and an outlet through which the introduced detergent is discharged; and a mixing space formed therein to receive the wash water from the tub and directly connected to the outlet of the detergent pump to receive the discharged detergent from the detergent pump; a drain pump connected to the tub to drain the wash water contained in the tub out of the cabinet; a circulation pump connected to the tub to circulate the wash water; a first circulation path to guide the wash water from the tub to the circulation pump; a second circulation path to guide the wash water from the circulation pump to the tub so that the wash water circulates by flowing from the tub to the circulation pump through the first circulation path and flowing back to the tub from the circulation pump through the second circulation path; and a processor configured to selectively operate the drain pump or the circulation pump, wherein the first circulation path includes a mixing path on which the outlet of the detergent pump is mounted and having a mixing space formed therein to receive the wash water from the tub and directly connected to the outlet of the detergent pump to receive the discharged detergent from the detergent pump.
 19. The clothes treating apparatus of claim 18, wherein the processor is further configured to operate the detergent pump in response to the operating of the circulation pump.
 20. The clothes treating apparatus of claim 19, wherein the processor is further configured to turn off the circulation pump and turn on the drain pump to drain the wash water contained in the tub.
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